

BMJ Open The effect of minimum volume standards in hospitals (MIVOS): a systematic review

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To cite: Scharfe J, Pfisterer-Heise S, Pachanov A, *et al.* The effect of minimum volume standards in hospitals (MIVOS): a systematic review. *BMJ Open* 2025;**15**:e090152. doi:10.1136/bmjopen-2024-090152

► Prepublication history and additional supplemental material for this paper are available online. To view these files, please visit the journal online (<https://doi.org/10.1136/bmjopen-2024-090152>).

Received 18 June 2024

Accepted 31 March 2025



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ABSTRACT

Objectives The relationship between the volume of medical procedures conducted in hospitals and the resulting health outcomes has been described for various surgical and non-surgical medical interventions. As a policy response to this, several countries have implemented minimum volume standards. However, there is currently a lack of systematically compiled evidence assessing their impact. To close this research gap, we conducted a systematic review on the effects of minimum volume standards in hospitals.

Design Systematic review using the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach.

Data sources MEDLINE, PubMed Central and Bookshelf (PubMed), EMBASE (Elsevier), CENTRAL (Cochrane Library), CINHAL (EBSCO), EconLIT (EBSCO), PDQ-Evidence for Informed Health Policymaking, Health Systems Evidence and three trial registries were searched until June 2023. Additionally, manual searches were conducted.

Eligibility criteria for selecting studies For inclusion in this review, studies must investigate the effects of minimum volume standards. We accepted all categories of outcomes. Following the Cochrane Effective Practice and Organisation of Care (EPoC) criteria, only a strict set of study designs, namely randomised controlled trials, non-randomised controlled trials, controlled before-after studies and interrupted time series studies, were included. No restrictions were placed on language, publication date or publication status.

Data extraction and synthesis Two reviewers independently screened titles and abstracts for eligibility, reviewed the full texts and performed data extraction of the included articles. Risk of bias was assessed using the 'Risk Of Bias In Non-Randomised Studies—of Interventions' (ROBINS-I) tool. Certainty of evidence was rated using the GRADE approach. For data synthesis and statistical analyses, we adhered to the EPoC guidance.

Results We included nine studies into our systematic review. Overall, the outcomes reported in the studies are heterogeneous, which did not allow for a meta-analysis, resulting in a narrative analysis of the found evidence. While travel distance increased and length of stay decreased, other reported outcomes such as complications or reoperations were not affected by the introduction of minimum volume standards. Overall, the risk of bias was considered serious for results on outcomes of seven

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ Systematically collated evidence on the effects of minimum volume standards in hospitals.
- ⇒ Inclusion of studies with robust study design instead of focusing on mere associations in observational studies.
- ⇒ Even after re-analysis, only nine studies could be included in this systematic review, of which results on outcomes of seven studies are considered to be at serious risk of bias. The certainty of evidence is very low for mortality and low for complications, reoperations, travel distance and length of stay.
- ⇒ There is a lack of research regarding the effects of minimum volume standards with robust study designs, such as randomised controlled trials, controlled before-after studies and interrupted time series studies with at least six data points (three before and three after the intervention).

out of the nine included studies and moderate for the remaining two. The certainty of evidence was rated low for complications, reoperations, length of stay and travel distance and very low for mortality (in-hospital; 30 days).

Conclusion This systematic review does not provide conclusive evidence on the effects of minimum volume standards in hospitals regarding any outcomes. The certainty of evidence for mortality (in-hospital; 30 days) is very low and low for complications, reoperations and travel distance. The results are based on a relatively small number of studies for most outcomes. There is a need for studies researching the effects of minimum volume standards based on a robust study design.

PROSPERO registration number CRD42022318883.

INTRODUCTION

Minimum volume standards are cut-off thresholds for hospitals allowing only those hospitals above a defined threshold to perform a given medical procedure in the future. As a policy tool, minimum volume standards aim at concentrating the procedures for which they are in effect in hospitals with higher volumes.¹ This objective is based on evidence of the volume-outcome

relationship according to which higher case volumes are associated with improved outcomes.^{2 3} In fact, evidence for a volume-outcome relationship has been accumulated for various medical interventions including surgeries, dialysis, perinatal as well as intensive care.⁴⁻¹⁴

Based on these findings, minimum volume standards have been introduced in numerous countries, particularly for highly specialised surgical procedures.^{15 16} Still, criticism remains. First, some studies have raised doubts about the underlying proposed volume-outcome relationship.^{17 18} Furthermore, uniform international guidance for minimum volume standards is seldom established, resulting, for example, in different cut-off thresholds and different regulatory approaches across different countries.¹⁵ Last, different effects of minimum volume standards have been discussed in the literature, including adverse effects such as an expansion of indications as well as a threat to emergency care.¹⁹

At the same time, there is no reported systematic review collating evidence on the effects of minimum volume standards in hospitals. In Germany, a rapid review was published in 2012 by the Institute for Quality and Efficiency in Healthcare (IQWiG). In total, 10 studies were included in this report. The majority of them were before-after studies with only one time point of data collection before and one after the implementation of minimum volume standards. As two time points in total, however, do not allow for the calculation of time trends, the authors' conclusion was that 'all studies showed a design-related potential for bias as well as methodological shortcomings that limited the validity of the results'.²⁰⁻²²

Hence, despite the ongoing debate about the advantages and disadvantages of minimum volume standards, there has been no systematic review on the effects of minimum volume standards in hospitals. This systematic review aims to address this research gap.

METHODS

Reporting of this systematic review follows the guidelines set forth in the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 Statement²³ (see online supplemental material 2). A priori, the systematic review was registered in the Prospective Register of Systematic Reviews (PROSPERO registration number CRD42022318883) and a protocol was published.²² When recycling text from the protocol, we followed the recommendations provided by the Text Recycling Research Project.²⁴

Eligibility criteria

Population

We included studies dealing with patients irrespective of their condition or the medical treatment received.

Intervention

The intervention was minimum volume standards in hospitals, defined as a minimum of specific healthcare

procedures (or, specifically, treatments), usually defined by a cut-off, in a given time frame and region. There were no restrictions with respect to the regulator (eg, state authority, regional authority or professional association), the year of implementation, selected standards or cut-off points and consequences in case of non-compliance (eg, non-reimbursement for the performed procedure). We exclusively took into account enforceable policies and disregarded mere recommendations, such as those from medical societies. As an amendment to the published protocol, we also accepted policies which combined minimum volume standards with certain co-interventions, that is, training of medical staff.

Comparator

The comparator was no minimum volume standard in hospitals.

Outcomes

We considered all outcomes. This includes patient-related outcomes as well as health system-related outcomes.

Design of primary studies

We included the following study designs:

- ▶ (Cluster) randomised controlled trials ((C)RCTs)
- ▶ Non-randomised controlled trials (nRCTs)
- ▶ Controlled before-after (CBA) studies.
- ▶ Interrupted time series (ITS) with a clearly defined time point of intervention introduction and at least three data points both before and after the intervention.

(C)RCTs are often not available to address questions about health system interventions. Therefore, we included other study designs based on the recommendations by the Cochrane Effective Practice and Organisation of Care (EPOC) group. The study design was determined using the Cochrane EPOC group algorithm.²⁵

No restrictions were placed on language, publication date or publication status of the studies. In case an article published in a language other than English or German was deemed eligible for inclusion based on the abstract screening, we had it translated into German.

Information sources and search strategies

The search strategy was developed by the research team in collaboration with an experienced librarian and checked by an expert according to the Peer Review of Electronic Search Strategies guideline.²⁶ The search strategies are provided in online supplemental material 3.

A comprehensive search of nine major databases/search engines from their date of inception to 15 June 2023 was performed. The following databases were searched to identify relevant studies: MEDLINE, PubMed Central and Bookshelf (via PubMed), Embase (via EMBASE), CENTRAL (via Cochrane Library), CINHAL (via EBSCO), EconLit (via EBSCO), PDQ-Evidence for Informed Health Policymaking and Health Systems Evidence. We also searched manually for additional studies by cross-checking the reference lists of relevant

systematic reviews and all included primary studies. Furthermore, we searched the following trial registries: clinicaltrials.gov, German Clinical Study Register and International Clinical Trials Registry Platform. We also contacted experts for additional studies and conducted a hand search of available abstracts from conference reports.

Data management and study selection

All potentially relevant hits were imported to a reference management software (Rayyan). Duplicate publications were removed. Two reviewers (JS, SP-H) independently screened titles and abstracts of all identified articles. We retrieved the full texts of all potentially relevant articles. Full-text articles were reviewed in detail regarding inclusion criteria by three reviewers (AP, JS, SP-H) in pairs of two authors, independently. In case of disagreement, eligibility was determined by discussion and consensus.

Data collection and quality assessment

A standardised data extraction tool was developed in Excel and calibrated within the team. Using a random sample of three of the included studies, the data extraction form was pilot-tested and revised, as necessary. Data extraction begun when high inter-rater reliability ($\kappa \geq 0.60$) was achieved.²⁷

Two authors (AP, JS) independently performed data extraction of the included articles. Subsequently, both reviewers checked each other's versions for completeness and accuracy. Discrepancies were resolved by discussion. We extracted data on the following items: sample size; study design; year(s) of data collection; country/region; data source; procedure or treatment (disease); definition of minimum volume standard; outcomes; (unadjusted and adjusted) effect measures with corresponding confidence intervals and/or p values; statistical models; and adjusting variables.

For quality assessment, the ROBINS-I tool, with additions for CBA and ITS study designs,²⁸ was independently applied by two reviewers (AP, JS). For both data collection and quality assessment, any disagreement was resolved by discussion and consensus.

Data synthesis

For data synthesis and statistical analyses, we followed the guidance published by the Cochrane EPOC group.²⁹ For dichotomous outcomes, we used the risk ratio (RR) obtained from statistical analyses adjusting for baseline differences (such as Poisson regressions or logistic regressions) or the ratio of RRs (ie, the RR postintervention/RR preintervention), if possible. For continuous outcomes, we used the absolute change obtained from a statistical analysis that was adjusted for baseline differences (eg, regression models, mixed models or hierarchical models). Alternatively, we used the relative change adjusted for baseline differences in the outcome measures. For ITS studies, if possible, we relied on the results either obtained by a regression including time trends before

and after the intervention adjusting for autocorrelation and any periodic changes or auto-regressive integrated moving average (ARIMA) models.

If papers with a before-after design did not provide an appropriate analysis or reporting of results to be included in this systematic review, but presented the needed data points in a readable graph or in a table, we re-analysed the data ourselves. We used segmented regression models, specifically ARIMA models with two external variables, to estimate level change and slope parameters associated with the intervention.³⁰ While level change reflects the sudden, sustained shift of the time series at the time of intervention, the slope parameter represents the change in the slope of the time series after intervention. We used the 'auto.arima' function from the R-package 'forecast' to select the most appropriate ARIMA model based on Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). 95% CIs were calculated for all effect measures. The adequacy of the model was assessed using residual plots and Ljung-Box tests to ensure that the model adequately captures the structure of the data. For data synthesis, we used R.

Certainty assessment

The certainty of evidence was evaluated by two reviewers (JS, AP) independently, based on the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach^{31–33} using the GRADEpro GDT software. We chose mortality (in-hospital; 30 days), travel distance, complications, reoperations and length of stay as outcomes of importance for assessing the effect of minimum volume standards interventions in hospitals. As strongly recommended in the GRADE handbook,³¹ researchers must base the choice of outcomes on what is important regarding the focus of the systematic review, not on what outcomes are measured and for which evidence is available. Acknowledging the absence of evidence for a significant outcome is preferable to disregarding it. Therefore, we also chose to add two outcomes from the developed Core Outcome Set for studies on centralisation of healthcare services,³⁴ 'routine/experience of surgeons' and 'quality of the medical indication'. Among all the groups of interest-holders investigated, including patient representatives, representatives of statutory health insurance funds, representatives of medical societies and health services researchers, these outcomes received the highest approval rates, which pertains to the significance of outcomes concerning centralisation.

Advisory board

At the beginning of our systematic review, we established a project advisory board comprising representatives from health services research and decision-makers. The primary objectives of this advisory board were to offer technical guidance on content-related matters and to ensure the timely dissemination of research results.

Patient and public involvement

Patients or the public were not involved in the design, conduct or reporting of our research. Both patient

representatives and the public were involved in the dissemination of our research findings.

RESULTS

Study identification and selection

In total, we located 12 878 records through electronic databases and trial registries, while an additional 642 records were discovered through reference lists of included articles, forward citation searches, and by contacting authors directly. After examining 241 full-text reports, 9 studies^{35–43} were subsequently included (see PRISMA flow diagram in online supplemental material 4).

Study and patient characteristics

Three studies used data from the Netherlands,^{36 40 43} each two from Canada^{35 41} and Switzerland,^{38 39} followed by one study from the USA³⁷ and Spain.⁴² All data were sourced from administrative databases. Across all nine included studies, the total number of patients was 334 197.

The studies focused on various surgical procedures subject to minimum volume standards, including lung cancer surgeries,³⁵ bariatric surgeries,³⁷ oesophagectomies,³⁶ liver resections,³⁸ gastrectomies^{40 43} and abdominal aortic aneurysm repairs.⁴² For pancreatic resections, this systematic review included two studies originating in two different countries with differing minimum volume standards.^{39 41} All study characteristics including study designs, countries, patient characteristics and diseases are presented in online supplemental material 1. Furthermore, the characteristics of the various minimum volume standards across the countries included in this systematic review can be found in online supplemental material 5.

Study designs

Two of the studies included had ITS designs,^{35 41} and one study had a CBA study design.³⁷ Moreover, four studies had an uncontrolled before-after study design,^{38 40 42 43} and two studies had a cohort study design.^{36 39} In the case that in these studies enough data points for an outcome were presented in a readable graph or a table, the available data was re-analysed using an ITS design as described above. In total, outcomes measured and reported in six studies (67%) were re-analysed. All outcomes reported in the included studies, along with information on whether these outcomes could be re-analysed, are presented in online supplemental material 1.

Study results

The studies covered a broad range of outcomes. Most studies focused on patient-related outcomes, while health system-related outcomes were seldom measured. We chose the following five outcomes, measured and reported in six studies,^{35–37 40–42} as outcomes of importance for assessing the effect of minimum volume standards interventions in hospitals: *mortality*, *travel distance*, *complications*, *reoperations* and *length of stay*. Table 1 presents

the characteristics of interventions and findings for all outcomes of importance.

Mortality

Three studies measured the impact of minimum volume standards in hospitals on mortality (in-hospital; 30 days),^{35 41 42} while one study focused on 6 month mortality.³⁶ Only one study reported a significant reduction of the short-term mortality rate.⁴² While the effect size for mortality varies among the individual studies, the direction of the effect is consistent: since the introduction of minimum volume standards, the mortality rate for the procedures included in the respective studies has decreased. The evidence collated in this systematic review is very uncertain about the effect of minimum volume standards on short-term mortality (in-hospital; 30 days) (very low certainty of evidence).

Travel distance

The distance travelled to treating hospitals increased significantly in both studies.^{35 40} Minimum volume standards in hospitals may result in a large increase in travel distance (low certainty of evidence).

Complications and reoperations

Complications^{35 37} and reoperations^{35 37} were each reported by two studies, respectively. The number of complications and reoperations was not affected by the introduction of minimum volume standards in hospitals. The evidence suggests that minimum volume standards in hospitals may result in little to no difference in complication or reoperation rates (low certainty of evidence).

Length of stay

The mean length of stay decreased significantly in one study.³⁵ Minimum volume standards in hospitals may result in a reduction of the mean length of stay (low certainty of evidence).

Routine/experience of surgeons and quality of the medical indication

There were no studies investigating the effects of minimum volume standards on routine/experience of surgeons and quality of medical indication.

Risk of bias and certainty assessment

The risk of bias was moderate for results on outcomes for two studies and serious for seven studies (see online supplemental material 6).

Across all outcomes examined in the studies, the risk of bias pertaining to the selection of both exposed and non-exposed populations, as well as classification of intervention, deviation from intended intervention, missing data and measurement of outcomes was determined to be low. For two studies, the possibility of bias due to selection of reported results was considered to be low.^{35 37} Seven studies were identified as probably having a serious risk of bias mostly due to potential confounding and the selection of reported results.^{36 38–43}

Table 1 Characteristics of interventions and findings for outcomes of importance (measured and reported in six studies)

Authors	Study design	Country	Intervention	Outcome(s)	Findings
Bendtsak <i>et al.</i> 2017 ³⁵	ITS	Canada	Regionalisation of lung cancer surgeries to 14 designated high-volume hospitals. ► MVR Level 1 centres: minimum 150 anatomic pulmonary resections/year. Level 2 centres: minimum 50 anatomic pulmonary resections/year. ► Other criteria: certified surgeons, 24-hour operating room availability, dedicated thoracic units, cancer centre affiliations, and an interdisciplinary team	Mortality (in hospital; 30 days) Complications Reoperations Travel distance Length of stay	OR=0.98; 95% CI 0.95 to 1.01) OR=1.03; 95% CI 0.96 to 1.10 OR=1.00; 95% CI 0.90 to 1.10) Increase by 4%/year (95% CI 0% to 8%) Decrease by 7%/year (95% CI 5% to 9%)
Dikken <i>et al.</i> 2012 ³⁶	ITS*	Netherlands	MVR: ► As of 2006 minimum 10 oesophagectomies/year ► As of 2011 minimum 20 oesophagectomies/year	Mortality (6 month)	Step†: 0.13; 95% CI –2.81 to 3.06 Ramp†: 0.08; 95% CI –0.99 to 1.15
Dimick <i>et al.</i> 2013 ³⁷	CBA	USA	Centres for Medicare & Medicaid Services' restriction of bariatric surgery to centres of excellence. Accreditation required: ► MVR of >125 bariatric surgeries/year ► Other structural measures and processes of care	Complications	RR=1.14; 95% CI 0.95 to 1.33 RR=0.90; 95% CI 0.64 to 1.17
Luijten <i>et al.</i> 2022 ⁴⁰	ITS*	Netherlands	Centralisation of surgical treatment for oesophageal, pancreatic and gastric cancer ► MVR: ► 10 resections/year initially for oesophageal and pancreatic cancer (2005–2006) and gastric cancer (2012) ► 20 resections/year in 2011 for oesophageal and pancreatic cancer and in 2013 for gastric cancer	Median travel distance	Step†: 1.36; 95% CI –2.25 to 4.97 Ramp†: Increase by 1.77 km/year (95% CI 0.15 to 3.38)
Simunovic <i>et al.</i> 2010 ⁴¹	ITS	Canada	Quality Improvement Interventions in Ontario in 1999: ► publication in a medical journal of an Ontario volume-outcomes study ► report on standards for major pancreas surgery – Minimum volume standard of 10 major pancreatic surgeries/year – Operative mortality is ≤5% – audit and feedback exercise with surgeons	Mortality (operative)	–0.08, p=0.53
Tripodi <i>et al.</i> 2020 ⁴²	ITS*	Spain	Centralisation of abdominal aortic aneurysm (AAA) repairs in 10 selected hospitals: ► MVR of >30 AAA repairs/year	Mortality (in hospital; within 30 days): rAAA OpR Mortality (in hospital; within 30 days): rAAA EVAR Mortality (in hospital; within 30 days): iAAA OpR Mortality (in hospital; within 30 days): iAAA EVAR	Step†: 3.34; 95% CI –12.00 to 18.68 Ramp†: –3.44; 16; 95% CI –10.30 to 3.43 Step†: 0.16; 95% CI –24.00 to 24.31 Ramp†: –2.87; 95% CI –13.67 to 7.94 Step†: Decrease by 6.06 deaths/year (95% CI –10.55 to –1.57) Ramp†: 0.43; 95% CI –1.58 to 2.44 Step†: –2.09; 95% CI –4.36 to 0.18 Ramp†: 0.70; 95% CI –0.32 to 1.71

Continued

Table 1 Continued				
Authors	Study design	Country	Intervention	Outcome(s)
*Re-analysed studies.				
†Step change (also called a level shift): A sudden, sustained change where the time series is shifted either up or down by a given value immediately following the intervention. The step change variable takes the value of 0 prior to the start of the intervention, and one afterwards. ³⁰				
‡Ramp: A change in slope that occurs immediately after the intervention. The ramp variable takes the value of 0 prior (horizontal line) to the start of the intervention and increases by one unit after the date of the intervention. ³⁰				
CBA, controlled before-after ; EVAR, endovascular aortic repair; iAAA, intact abdominal aortic aneurysm; ITS, interrupted time series; MVR, minimum volume standards; OpR, open repair; rAAA, ruptured abdominal aortic aneurysm; RR, relative risk.				

GRADE

The overall rating for certainty in estimates was very low for the outcome mortality (in-hospital; 30 days). The outcomes complications, reoperations, length of stay and travel distance were rated as of low certainty of evidence. We also included two outcomes resulting from our development of a core outcome set for studies on centralisation of healthcare services: routine/experience of surgeons and quality of the medical indication. The GRADE assessments are reported in the summary of findings table (table 2).

DISCUSSION

This article presents a rigorous systematic review, including a re-analysis of observational studies, to evaluate the effects of minimum volume standards in hospitals. Due to the scarcity of the articles and the heterogeneity of the data, we conducted structured syntheses without meta-analyses. This led to the conclusion that there is not enough evidence regarding the effects of minimum volume standards in hospitals, yet.

The vast majority of outcomes included in this study are patient-related. As minimum volume standards impact the entire health system and can have a number of effects both at the process level and the health system level, this is a noteworthy result. This is also the reason why we included two outcomes of our currently developed core outcome set for studies on centralisation of healthcare services into our summary of findings table (table 2).

While there is a consistent trend of decreasing mortality rates in all reported studies following the introduction of minimum volume standards, it is important to note that this effect is statistically significant only in subgroup analyses for endovascular aortic repair⁴² and lung cancer surgery for patients aged 70+ years.³⁵ As expected, the distance between the patient's residence and the treating hospital statistically significantly increased during the process of centralisation through the implementation of minimum volume standards in both studies.^{35 40}

Overall, considering the evidence for the volume-outcome relationship in hospitals,^{44 45} the lack of findings in our systematic review with respect to complication and mortality rates is noteworthy. More precisely, evidence whether the introduction of minimum volume standards based on the volume-outcome relationship, leads to improved complication and mortality rates, is still lacking. Notably, volume-outcome studies employ varying thresholds, a critical aspect contributing to the observed outcome in our review. The determination of minimum volume cut-offs poses a formidable challenge and lacks empirical grounding. Bearing in mind that the certainty of the evidence is considered low or very low, a possible explanation may be that the thresholds for the minimum volume standards collated in this systematic review are still too low.

There is no international consensus on what level of minimum volume standards for certain medical procedures yields the best results in terms of treatment quality.

Table 2 Summary of findings and certainty of evidence (GRADE)

Number of studies	No. of participants (study designs)	Effect	Certainty	Importance
			Certainty rating Reason for rating	
Mortality (in-hospital; 30 days)				
3 studies (Bendzsak <i>et al.</i> ³⁵ 2017, Simunovic <i>et al.</i> ⁴¹ 2010 Tripodi <i>et al.</i> ⁴² 2020]	n=23,842 (ITS)	While the effect size for mortality varies among the individual studies, the direction of the effect is consistent: since the introduction of minimum volume standards, the mortality rate for the procedures included in the respective studies has decreased. The evidence collated in this systematic review is very uncertain about the effect of minimum volume standards on short-term mortality (in-hospital; 30 days).	⊕○○○ VERY LOW –2 for risk of bias –1 for inconsistency –2 for indirectness –1 for imprecision	CRITICAL
Complications				
2 studies (Bendzsak <i>et al.</i> ³⁵ 2017 Dimick <i>et al.</i> ³⁷ 2013]	n=289,893 (CBA, ITS)	The evidence suggests that minimum volume standards in hospitals may result in little to no difference in complication rates.	⊕⊕○○ LOW –1 for risk of bias –1 for imprecision	CRITICAL
Reoperations				
2 studies (Bendzsak <i>et al.</i> ³⁵ 2017 Dimick <i>et al.</i> ³⁷ 2013]	n=289,893 (CBA, ITS)	The evidence suggests that minimum volume standards in hospitals may result in little to no difference in reoperation rates.	⊕⊕○○ LOW –1 for risk of bias –1 for imprecision	CRITICAL
Travel distance				
2 studies (Bendzsak <i>et al.</i> ³⁵ 2017 Luijten <i>et al.</i> ⁴⁰ 2021]	n=23,384 (ITS)	The distance travelled to treating hospitals increased significantly in both studies. Minimum volume standards in hospitals may result in a large increase in travel distance.	⊕⊕○○ LOW –2 for risk of bias	IMPORTANT
Routine/experience of the surgeons				
0 studies				IMPORTANT
Quality of the medical indication				
0 studies				IMPORTANT
Length of stay				
1 study ^{47–53}	n=16,641 (ITS)	The mean length of stay decreased significantly in one study. Minimum volume standards in hospitals may result in a reduction of the mean length of stay.	⊕⊕○○ LOW –1 for risk of bias, –1 for imprecision	NOT IMPORTANT
CBA, controlled before-after; GRADE, Grading of Recommendations, Assessment, Development and Evaluation; ITS, interrupted time series.				

For example, this systematic review included two studies originating in two different countries with differing minimum volume standards.^{39 41} Additionally, minimum volume standards are often introduced together with other quality-enhancing measures in medical care in hospitals. Here, the table 'Minimum Volume Standards' in online supplemental material 5 provides a further overview of which additional measures were implemented simultaneously with the introduction of the respective minimum volume standards. This makes it difficult to attribute effects to the minimum volume standards intervention.

It is both surprising and concerning that, in all studies in which the adherence to minimum volume standards in hospitals was assessed, a considerable number of procedures associated with minimum volume standards were still conducted in hospitals with case numbers well below the minimum volume threshold.^{35 39–41 43} Furthermore, it seems remarkable that there is a considerable amount of volume-outcome studies, while there is a limited focus on the effects of minimum volume standards, even though these regulations are widely employed in various countries to centralise healthcare services.¹⁵ From this already comparatively small

number of studies on minimum volume standards, even fewer have been conducted with a robust study design.

This finding is in line with the results of a systematic review on minimum volume standards in day surgery.^{20 46} Although the authors a priori had not put any limitations on the study design, they finally concluded: ‘The retrospective study design can at best show correlations between surgeon/hospital volumes and day surgery outcomes’.⁴⁶ However, to evaluate the effects of minimum volume standards in hospitals, study designs as recommended by EPOC should be employed. Particularly, as (C)RCTs and nRCTs cannot be conducted in the context of minimum volume standards, controlled before-after studies or interrupted time series studies with at least three data points prior and post-intervention are needed. Although both study designs are increasingly discussed, they remain relatively unknown and are therefore not widely used. CBA studies are more challenging to design, as they require a comparable control group. In contrast, ITS designs are more feasible in many cases; however, too few observation points are often used.²¹ Given that most studies rely on registry or administrative data, increasing the number of observation points is generally feasible and could enhance the use of ITS analyses.

LIMITATIONS

While this systematic review was conducted with care, it is not without limitations. Results on outcomes of seven out of nine studies are considered to be at serious risk of bias. The certainty of evidence is very low for mortality (in-hospital; 30 days) and low for complications, reoperations, length of stay and travel distance. The results are based on a relatively small number of studies for most outcomes. This is mostly due to our restriction to only include studies with a robust study design (eg, CBA or ITS studies). Only three of nine (33%) included studies had a thorough study design that could be clearly identified as either a CBA³⁷ or an ITS.^{35 41} Despite the efforts to include grey literature, the search for conference abstracts and contacting authors, there is still a potential for publication bias. Thus, reported outcomes should be interpreted with caution. Nevertheless, our study provides valuable insights into the effects of minimum volume standards and the call for studies researching those effects with a robust study design.

CONCLUSION

There is strong evidence for the existence of a volume-outcome relationship in healthcare, which provides the rationale for the implementation of minimum volume standards in hospitals. Those standards have been widely implemented, if not with the same cut-off values in various countries worldwide.¹⁵ Yet, there is a considerable lack of evidence for the effects of the minimum volume standards already in place. To ascertain the effectiveness of the implemented minimum volume standards, we recommend conducting evaluations of existing minimum volume standards as part of the respective

implementation process. Those evaluations should be based on robust study designs, such as ITS or CBA study designs.

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Funding This work was supported by the German Federal Ministry of Education and Research (Grant: 01KG2107). The German Federal Ministry of Education and Research did not take part in developing this study. The publication was funded by the Brandenburg Medical School publication fund supported by the Ministry of Science, Research and Cultural Affairs of the State of Brandenburg.

Competing interests None declared.

Patient and public involvement Patients and/or the public were involved in the design, conduct, reporting or dissemination plans of this research. Refer to the Methods section for further details.

Patient consent for publication Not applicable.

Ethics approval Not applicable.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request. The data that support the findings of this study are available from the corresponding author (JS) upon reasonable request.

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